Time-dependent Schrödinger equation for the multichannel ZEPE problem in the ^{85}Rb halo: full-TDSE and hybrid recipes

Goal and warning

The goal is to compute the channel-resolved dissociation spectrum

$$\frac{dP^{(M_F)}}{dE_f} \tag{1}$$

for the ^{85}Rb Feshbach halo at $B = 155.04$ G driven by a finite-bandwidth single-helicity RF pulse. This note gives two numerically distinct routes:

1. **Full multichannel TDSE:** include the blocks $M_F = -5, M_F = -4, M_F = -3, M_F = -2$ explicitly and propagate them together. This is conceptually clean, but the $M_F = -5$ virtual block can require a large spectral cutoff if it is represented by box eigenstates.
2. **Hybrid TDSE + resolvent:** propagate the open/observable blocks in time and eliminate the far-detuned $M_F = -5$ block by an inhomogeneous resolvent. This is usually the recommended production method, but it is a hybrid method, not a literal full TDSE over all blocks.

The clean ZEPE protocol requires **single-helicity σ^+ RF**. A linearly polarized transverse RF field contains both helicities and can reintroduce mixed-helicity paths. Therefore this recipe is written for the σ^+ component. Linear polarization can be tested later as an imperfection, not used as the baseline isolation protocol.

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1. System, Hamiltonian, and fixed numerical scales

1.1. Physical parameters

Two ground-state ^{85}Rb atoms are treated in relative s-wave motion. The working square-well model is calibrated to the ^{85}Rb halo at $B = 155.04$ G.

Symbol	Recommended value	Meaning
$m_{^{85}\mathrm{Rb}}$	$84.911789738\,m_u$	atomic mass
μ	$77{,}392\,m_e$	reduced mass
B	155.04 G	static bias field
E_h/h	-10.108 kHz	target halo energy
$1/\kappa$	$\simeq 2.05\times 10^3\,a_0$	halo size
r_0	$82.10\,a_0$	square-well range
V_T	9.6930959 GHz	triplet depth after sign correction
V_S	$1.02V_T$	singlet depth
L	$5\times 10^5\,a_0$	large box for open near-threshold states
τ_0	25–50 μs	useful pulse-ramp scale
$\omega/(2\pi)$	50–100 kHz	clean ZEPE carrier window

Relative to the $M_F\!=\!-4$ open-channel threshold, the relevant two-channel blocks have asymptotic thresholds

$$M_F\!=\!-4:\quad \{0,\ 2605.76\}\ \text{MHz}, \tag{2}$$

$$M_F\!=\!-3:\quad \{-77.42,\ 2532.24\}\ \text{MHz}, \tag{3}$$

$$M_F\!=\!-5:\quad \{2683.18,\ 5448.24\}\ \text{MHz}, \tag{4}$$

$$M_F\!=\!-2:\quad \text{computed from the same Breit–Rabi basis.} \tag{5}$$

The lower $M_F\!=\!-5$ threshold is 2.68 GHz above the entrance threshold. This is the central reason that $M_F\!=\!-5$ is a virtual, far-detuned block for all kHz RF carriers.

1.2. Field-free Hamiltonian

For each fixed-M_F block,

$$\hat{H}_0 = -\frac{\hbar^2}{2\mu}\frac{d^2}{dr^2}\mathbb{1} + \hat{V}_{\text{short}}(r) + \hat{H}_{\text{hf+Z}}(B). \tag{6}$$

The one-atom hyperfine-Zeeman Hamiltonian is diagonalized in the $|m_I, m_S\rangle$ basis,

$$\hat{H}^{(1)}_{\text{hf+Z}} = a_{\text{hf}}\hat{\mathbf{I}}\cdot\hat{\mathbf{S}} + \mu_B B(g_S\hat{S}_z + g_I\hat{I}_z), \tag{7}$$

then two-atom symmetrized channel states are formed at fixed M_F.

The square-well spin potential is

$$\hat{V}_{\text{short}} = -V_S\hat{P}_s - V_T\hat{P}_t, \tag{8}$$

with

$$\hat{P}_s = \frac{1}{4}\mathbb{1} - \hat{\mathbf{s}}_1\cdot\hat{\mathbf{s}}_2, \qquad \hat{P}_t = \frac{3}{4}\mathbb{1} + \hat{\mathbf{s}}_1\cdot\hat{\mathbf{s}}_2. \tag{9}$$

Therefore

$$\boxed{\hat{V}_{\text{short}}(r) = \Theta(r_0 - r)\left[-\bar{V}\mathbb{1} + \Delta V\,\hat{\mathbf{s}}_1\cdot\hat{\mathbf{s}}_2\right]}, \qquad \bar{V} = \frac{V_S + 3V_T}{4}, \quad \Delta V = V_S - V_T. \tag{10}$$

The positive sign in front of $\Delta V\,\hat{\mathbf{s}}_1\cdot\hat{\mathbf{s}}_2$ is important.

The coupled radial eigenproblem is

$$-\frac{\hbar^2}{2\mu}u''_\alpha(r) + E^{\text{th}}_\alpha u_\alpha(r) + \sum_\beta V_{\alpha\beta}(r)u_\beta(r) = Eu_\alpha(r), \qquad u_\alpha(0) = 0. \tag{11}$$

For box eigenstates use $u_\alpha(L) = 0$. For true scattering states use asymptotic matching; for the closed resolvent use decaying Robin boundary conditions.

1.3. Single-helicity RF coupling

Use a circular σ^+ component, not a linear x polarization, for the clean isolation calculation. In the dressed two-atom basis define

$$\eta^{(+)}_{\alpha\beta} = \langle\alpha|\hat{S}_{1,+} + \hat{S}_{2,+}|\beta\rangle, \qquad \eta^{(-)} = (\eta^{(+)})^\dagger. \tag{12}$$

The Hermitian classical RF interaction may be written schematically as

$$\hat{V}_{\mathrm{RF}}(t) = \frac{\Omega_R}{2}\,\chi(t)\left[e^{-i\omega t}\hat{\eta}^{(+)} + e^{+i\omega t}\hat{\eta}^{(-)}\right], \qquad \Omega_R = \mu_B g_S B_{\mathrm{RF}}, \tag{13}$$

where Ω_R is an energy in atomic units if $\hbar = 1$ is used in the code. Absorption of a σ^+ photon raises M_F by one unit:

$$M_F = -4 \xrightarrow{\text{absorb}} M_F = -3, \qquad M_F = -3 \xrightarrow{\text{absorb}} M_F = -2. \tag{14}$$

Stimulated emission of a σ^+ photon lowers M_F by one unit:

$$M_F = -4 \xrightarrow{\text{emit}} M_F = -5. \tag{15}$$

These selection rules are the reason the main ZEPE branch, the closed virtual branch, and the absorb-absorb background can be channel-resolved.

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2. Route A: full multichannel TDSE

2.1. When to use it

Use the full TDSE when the goal is a direct, non-perturbative validation of the experimental protocol. It is the cleanest conceptual calculation because no closed block is eliminated. It should include at least

$$\boxed{M_F = -5, \quad M_F = -4, \quad M_F = -3, \quad M_F = -2.} \tag{16}$$

The roles are

$$\begin{aligned}
M_F = -4 &: \quad \text{initial halo and detected ZEPE final channel,} && (17)\\
M_F = -3 &: \quad \text{one-photon channel and absorb–emit intermediate,} && (18)\\
M_F = -2 &: \quad \text{absorb–absorb two-photon exit channel,} && (19)\\
M_F = -5 &: \quad \text{emit–absorb virtual branch back to } M_F = -4. && (20)
\end{aligned}$$

2.2. Why the calculation can be numerically heavy

The problem is not that one needs energies of order Hartree. One does not. The relevant closed-block scales are GHz, not atomic-unit energies:

$$1\ \text{GHz} = 1.52 \times 10^{-7}\ \text{Ha}, \qquad 10\ \text{Ha} \simeq 6.58 \times 10^{16}\ \text{Hz}. \tag{21}$$

Thus a cutoff of 10 Ha is physically meaningless here: it is more than seven orders of magnitude above the square-well and threshold scales.

The cost comes from two different sources:

1. A lab-frame time propagation would need to resolve GHz phases in the $M_F = -5$ block. This is avoided by using an interaction-picture eigenbasis.
2. A box-eigenstate representation of the $M_F = -5$ virtual resolvent may need many high-energy box modes for completeness. Short-range sources require high-k radial modes even though the physical intermediate energy is near the entrance threshold.

2.3. Recommended basis and energy cutoffs

Diagonalize each block separately, then pool the eigenstates. Use the interaction-picture amplitudes, not lab-frame grid propagation.

For the open/detected blocks $M_F = -4, M_F = -3, M_F = -2$, choose the cutoff to cover all physically relevant outgoing features:

$$E_{\rm cut}^{\rm open} \gtrsim \max\left[20E_b,\ 2\hbar\omega - E_b + 5\hbar/\tau_0\right] \tag{22}$$

above the corresponding lowest open threshold. For the reference $\hbar\omega = 8E_b$, this means the absorb-absorb feature sits at $15E_b$, so $E_{\rm cut}^{\rm open} = 20\text{–}30E_b$ is normally adequate.

For the closed $M_F = -5$ block, a brute-force spectral TDSE needs a much larger cutoff measured in GHz, not in E_b:

$$\boxed{E_{\rm cut}^{(-5)} = 5,\ 10,\ 20,\ 50\ \text{GHz} \quad \text{above the lower } M_F = -5 \text{ threshold, tested for convergence.}} \tag{23}$$

A practical starting point is 20 GHz; continue to 50 GHz if P_{ea}/P_{ae} or the coherent peak height is still moving. The criterion is not convergence of individual $M_F = -5$ populations, which are virtual and basis-dependent, but convergence of physical observables:

$$P_{\rm ZEPE}^{(-4)}, \qquad P_{aa}^{(-2)}, \qquad E_{\rm peak}^{(-4)}, \qquad \frac{P_{ea}}{P_{ae}}, \qquad \Delta_{\rm coh}. \tag{24}$$

Stop when these change by less than 1% under a simultaneous increase of $E_{\rm cut}^{(-5)}$, L, and the smoothing resolution.

2.4. Full TDSE matrix

Let $|i\rangle$ denote a box/scattering-discretized eigenstate in one of the four blocks. In the interaction picture,

$$|\Psi(t)\rangle = \sum_i b_i(t) e^{-iE_i t/\hbar} |i\rangle. \tag{25}$$

The amplitude equation is

$$\boxed{i\hbar \dot{b}_f(t) = \frac{\Omega_R}{2}\chi(t) \sum_i \left[d_{fi}^{(+)} e^{i(E_f - E_i - \hbar\omega)t/\hbar} + d_{fi}^{(-)} e^{i(E_f - E_i + \hbar\omega)t/\hbar}\right] b_i(t),} \tag{26}$$

where

$$d_{fi}^{(+)} = \langle f|\eta^{(+)}|i\rangle, \qquad d_{fi}^{(-)} = \langle f|\eta^{(-)}|i\rangle. \tag{27}$$

No rotating-wave approximation should be made, because the ZEPE pathways use one positive- and one negative-frequency interaction.

Initial condition:

$$b_h(0) = 1, \qquad b_i(0) = 0 \quad (i \neq h). \tag{28}$$

Recommended integrators: fourth-order commutator-free Magnus or adaptive RK4/5 in the interaction picture. A simple RK4 is acceptable only if convergence with respect to Δt is checked. A useful starting step is

$$\Delta t \lesssim \frac{2\pi}{30\omega}, \tag{29}$$

then halve Δt until the spectra change by less than 1%.

2.5. Readout

After the pulse, compute separate spectra in each final M_F block:

$$\frac{dP^{(M_F)}}{dE}(E) = \frac{1}{\sqrt{2\pi}\delta E} \sum_{f \in M_F} |b_f(T)|^2 \exp\left[-\frac{(E - E_f)^2}{2\delta E^2}\right]. \tag{30}$$

Use δE equal to one to two times the local level spacing near the peak. The main outputs are

$$\frac{dP^{(-4)}}{dE}: \quad \text{detected ZEPE channel,} \tag{31}$$

$$\frac{dP^{(-3)}}{dE}: \quad \text{one-photon channel,} \tag{32}$$

$$\frac{dP^{(-2)}}{dE}: \quad \text{absorb–absorb channel.} \tag{33}$$

The $M_F = -5$ block should not be interpreted as an outgoing channel in this experiment; it is virtual unless a real population is found, which would signal breakdown of the far-detuned approximation.

2.6. Full-TDSE convergence checklist

1. Halo binding: $E_h/h = -10.108 \pm 0.001$ kHz.
2. Halo weights: $P_{\text{open}} = 0.99981$, $P_{\text{closed}} = 1.901 \times 10^{-4}$.
3. Norm conservation: $|1 - \sum_i |b_i(t)|^2| < 10^{-6}$.
4. Time-step convergence: halve Δt and require $< 1\%$ change in the peak and integrated yields.
5. Open cutoff convergence: increase $E_{\text{cut}}^{\text{open}}$ from $20E_b$ to $30E_b$ or $40E_b$.
6. Closed cutoff convergence: increase $E_{\text{cut}}^{(-5)} = 5, 10, 20, 50$ GHz; do not use Hartree-scale cutoffs.
7. Box convergence: double L and rescale the smoothing width with the local level spacing.
8. Weak-field check: in the perturbative limit, $P_{\text{ZEPE}}^{(-4)} \propto B_{\text{RF}}^4$, $P_{1\gamma}^{(-3)} \propto B_{\text{RF}}^2$, and $P_{aa}^{(-2)} \propto B_{\text{RF}}^4$.

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3. Route B: hybrid open-channel TDSE plus closed-block resolvent

3.1. What the hybrid method is

The hybrid method propagates the experimentally relevant open blocks in time and eliminates the far-detuned $M_F = -5$ block with a resolvent. It should be called exactly that:

$$\boxed{\text{open-channel TDSE } (M_F = -4, M_F = -3, M_F = -2) \; + \; \text{closed } M_F = -5 \text{ resolvent correction.}} \tag{34}$$

It is not a literal full TDSE over all blocks. It is justified because

$$\Delta_{m5}/h \simeq 2.68 \text{ GHz} \gg \omega/(2\pi) \sim 50\text{–}100 \text{ kHz}, \qquad \Delta_{m5} \gg \hbar/\tau_0. \tag{35}$$

Thus $M_F = -5$ is only virtually occupied and can be eliminated perturbatively in the RF amplitude while being treated exactly in its own Hamiltonian.

3.2. Open-channel TDSE subspace

Use the pooled basis

$$\{|i\rangle\}_{\text{open}} = \{M_F\!=\!-4\} \cup \{M_F\!=\!-3\} \cup \{M_F\!=\!-2\}. \tag{36}$$

Including $M_F\!=\!-2$ is recommended. It lets the calculation explicitly show where the absorb-absorb background goes:

$$M_F\!=\!-4 \xrightarrow{\text{absorb}} M_F\!=\!-3 \xrightarrow{\text{absorb}} M_F\!=\!-2. \tag{37}$$

If the only target is the detected $M_F\!=\!-4$ ZEPE channel, $M_F\!=\!-2$ can be omitted as a speed test, but the production calculation should include it.

Propagate the same interaction-picture TDSE as in Route A, but only on this open subspace. The detected $M_F\!=\!-4$ amplitude from this propagation contains the open absorb-emit ZEPE branch through $M_F\!=\!-3$ and all non-perturbative open-subspace corrections.

3.3. Closed $M_F\!=\!-5$ resolvent equation

For each carrier frequency solve once

$$\boxed{(E_{\text{int}} - H_{-5})X(r) = S(r), \qquad S(r) = \eta^{(-)}u_h(r), \qquad E_{\text{int}} = E_h - \hbar\omega.} \tag{38}$$

Here $\eta^{(-)}$ is the emission vertex from $M_F = -4$ to $M_F = -5$. Discretize with the three-point block-tridiagonal scheme

$$AX_{j-1} + B_jX_j + AX_{j+1} = -S_j, \tag{39}$$

$$A = -\frac{1}{2\mu(\Delta r)^2}\nVdash, \qquad B_j = \frac{1}{\mu(\Delta r)^2}\nVdash + V_j^{(-5)} - E_{\text{int}}\nVdash. \tag{40}$$

Boundary conditions:

$$X_0 = 0, \qquad X_{N,\alpha} = e^{-\kappa_\alpha \Delta r}X_{N-1,\alpha}, \qquad \kappa_\alpha = \frac{\sqrt{2\mu(E_{\text{th},\alpha}^{(-5)} - E_{\text{int}})}}{\hbar}. \tag{41}$$

In atomic units, drop the explicit $\hbar$ in the last formula. Solve by block-Thomas in $O(N_{\text{grid}})$ time. A small channel dimension, here two channels, makes this very cheap and stable.

Recommended closed-grid parameters:

$$\Delta r = 0.01a_0 \quad \text{for production,} \qquad L_{-5} \text{ large enough that } X \text{ is negligible at the Robin boundary.} \tag{42}$$

Start with the same large radial extent used for the halo tail; then halve Δr or double L_{-5} only if the physical observables change.

3.4. Closed-channel amplitude added to the TDSE result

For each final $M_F\!=\!-4$ state,

$$\langle f|\eta^{(+)}X\rangle = \sum_\alpha \int_0^L u_f^{(\alpha)}(r)^*\,[\eta^{(+)}X]^{(\alpha)}(r)\,\mathrm{d}r. \tag{43}$$

The large-detuning pulse kernel is

$$J_{\text{LD}}^{(+,-)}(E_f) = -i\hbar\tau_0\sqrt{\frac{\pi}{2}}\exp\left[-\frac{(E_f - E_h)^2\tau_0^2}{8\hbar^2}\right]. \tag{44}$$

Then

$$\boxed{c_f^{ea} = \frac{\Omega_R^2}{4\hbar^2}\langle f|\eta^{(+)}X\rangle J_{\mathrm{LD}}^{(+,-)}(E_f).} \tag{45}$$

The final detected amplitude is

$$\boxed{b_f^{\mathrm{det}}(T) = b_f^{\mathrm{open\ TDSE}}(T) + c_f^{ea}.} \tag{46}$$

The phase convention in this addition is important; see the validation checks below.

3.5. Hybrid validation checks

The hybrid result should not be trusted unless these checks pass:

1. **Resolvent residual:**

$$\frac{\|(E_{\mathrm{int}} - H_{-5})X - S\|}{\|S\|} < 10^{-10} \quad \text{or at least below the FD truncation floor.} \tag{47}$$

2. **Born-limit scale:**

$$(|X|/|S|)\Delta_{m5} \simeq 1, \qquad \Delta_{m5} = E_{\mathrm{th},-5} - E_{\mathrm{int}}. \tag{48}$$

   A deviation at the percent level is acceptable and reflects inside-well dynamics.

3. **Weak-field phase calibration:** at small B_{RF}, compare $M_F = -4$ amplitudes from open-channel TDSE with explicit second-order perturbation theory for the ae branch. Magnitude and phase must agree.
4. **Closed-kernel calibration:** verify the large-detuning formula against the full Faddeeva kernel for the $M_F = -5$ bound spectrum, or against a controlled spectral calculation with the bound part subtracted from X to avoid double counting.
5. **No double counting:** the open TDSE must not include $M_F = -5$ if c_f^{ea} is added separately.
6. **Power laws:** in the perturbative regime, $P_{\mathrm{ZEPE}}^{(-4)} \propto B_{\mathrm{RF}}^4$ and $P_{1\gamma}^{(-3)} \propto B_{\mathrm{RF}}^2$.
7. **Operating-point check:** evaluate the hybrid method directly at $\tau_0 = 30\,\mu$s and $\hbar\omega = 8E_b$, not only at $\hbar\omega = E_b/2$. Report P_{ea}/P_{ae}, Δ_{coh}, and the peak shift.

3.6. Why this route is usually recommended

The $M_F = -5$ block is far detuned and only virtual. Representing its effect by box eigenstates can require GHz-scale cutoffs and delicate cancellations between bound-like and continuum-like states. The inhomogeneous solve computes exactly the object needed by the physics,

$$X = (E_{\mathrm{int}} - H_{-5})^{-1} S, \tag{49}$$

without ever constructing the ill-conditioned spectral sum. Therefore the hybrid method is the recommended production route unless a fully converged four-block TDSE is explicitly required.

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4. Reference operating point and expected outputs

The recommended experimental ZEPE point is

$$\boxed{\tau_0 = 30\,\mu\mathrm{s}, \qquad \hbar\omega = 8E_b, \qquad \mu_B g_S B_{\mathrm{RF}}/h \simeq 179\ \mathrm{kHz}.} \tag{50}$$

Expected square-well outputs:

$$\frac{dP^{(-4)}}{dE}: \quad \text{ZEPE peak at } E_{\text{peak}}/k_B \simeq 200 \text{ nK}, \tag{51}$$

$$P^{(-4)}_{\text{ZEPE}}: \quad \sim 5 \times 10^{-5} \text{ per molecule per shot in PT}, \tag{52}$$

$$P^{(-4)}_{\text{ZEPE}}: \quad \sim 3.5 \times 10^{-5} \text{ per molecule per shot in TDSE}, \tag{53}$$

$$\frac{dP^{(-3)}}{dE}: \quad \text{one-photon feature near } E = \hbar\omega - E_b \simeq 3.4\,\mu\text{K}, \tag{54}$$

$$P^{(-3)}_{1\gamma}: \quad \sim 5 \times 10^{-2}. \tag{55}$$

For the multichannel correction, report at this same point

$$\frac{P_{ea}}{P_{ae}}, \qquad \Delta_{\text{coh}} = \frac{P_{\text{coh}} - P_{ae}}{P_{ae}}, \qquad \frac{|E^{\text{coh}}_{\text{peak}} - E^{ae}_{\text{peak}}|}{E^{ae}_{\text{peak}}}. \tag{56}$$

These three numbers are the cleanest diagnostics of whether the realistic multichannel structure preserves the ZEPE observable.

5. Short summary

- Do not use a 10 Ha cutoff. If diagonalizing $M_F = -5$, think in GHz-scale cutoffs: 5–50 GHz above the $M_F = -5$ threshold.
- Full TDSE is conceptually clean but requires careful convergence of the $M_F = -5$ virtual resolvent in a spectral basis.
- The hybrid method is more stable: propagate $M_F = -4, M_F = -3, M_F = -2$ and eliminate $M_F = -5$ with an inhomogeneous resolvent.
- Use σ^+ polarization for the clean isolation calculation. Treat linear polarization only as an imperfection test.
- Validate phases before trusting coherent yield enhancements.